Functional Features Included in the Solution

Phase 5: Project Development Phase

1. Feature Inventory

The project provides a set of related capabilities centered on the itinerary workflow. Features should be counted as implemented capabilities only when their behaviour is present in the source and can be demonstrated. The following inventory describes the expected behaviour and its boundary.

2. Trip Input

The form collects starting city, destination, dates, total budget, traveller count, interests and travel style. These inputs establish the context for itinerary generation. Validation should prevent missing or malformed values from being sent as a valid request.

3. AI Itinerary Generation

The backend calls Groq using the configured model and requests a structured itinerary. The result includes a trip summary, days, activities, estimated total cost and travel tips. This is the central AI capability. It creates a planning draft rather than a guarantee that every activity, travel time or cost is correct.

4. Daily Plan and Estimated Costs

The itinerary is organized into days and activity periods to make the schedule easier to read. Estimated costs provide an initial budget reference, but they are not live quotes. A traveller should check current admission fees, transport fares, accommodation costs and local conditions before making decisions.

5. Saved Trips

The frontend lists saved trips, opens a selected trip and allows deletion. Supabase stores trip and itinerary data. These actions make the output reusable instead of requiring a new generation for every visit. They should be tested after reloading the page to confirm that the data is actually persisted.

6. Hotels and Restaurants

The application constructs hotel and restaurant searches in Google Maps. Restaurant searches can include food preference terms. This is useful as an external discovery step but does not mean the app has live place records, verified prices, ratings or dietary guarantees.

7. Print and Save as PDF

The browser print flow allows users to print the itinerary or save it as PDF. Output appearance depends on the browser and print settings. The feature should be tested with the complete itinerary to ensure content is not cut off.

8. Backend Health Check

The /health endpoint provides a basic way to check whether the backend service is responding. It does not by itself prove that the Groq provider and Supabase are available or that an itinerary can be generated.

9. Feature Acceptance Checklist

- Trip form accepts a valid set of preferences.
- Itinerary generation returns the expected structure.
- Daily activities and estimates are visible.
- Saved trips persist and can be retrieved.
- Deletion removes the intended saved record.
- Map-search links contain the selected destination and food preference.

- Print/save opens the browser print interface.
- Errors are handled when an external service fails.

No performance score or feature pass percentage is asserted here. Record the results from actual test runs before using the inventory as a completed acceptance report.